

Please submit your solutions to the following problems on Gradescope by **6pm** on the due date. Collaboration is encouraged, however, you must write up your solutions individually.

1) Taylor series. Finite difference formulas are often used to approximate derivatives from function samples on a grid. For example, the second-order centered difference approximation to the first derivative of a function f on an equally spaced grid with grid spacing $h > 0$ is

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

- (a) If $u(x)$ has three continuous derivatives, show that the centered difference formula approximates $u'(x)$ with accuracy $\mathcal{O}(h^2)$ as $h \rightarrow 0$. Derive an explicit upper bound for the approximation error using an appropriate Taylor polynomial.
- (b) How does the bound in part (a) change if $u(x)$ is only twice continuously differentiable?
- (c) Derive a fourth-order accurate centered difference formula to approximate $u'(x)$ from samples $u(x-2h), u(x-h), u(x), u(x+h), u(x+2h)$ with grid spacing $h > 0$. Here, “fourth-order” accurate means that your approximation should have accuracy $\mathcal{O}(h^4)$ as $h \rightarrow 0$ when $u(x)$ has five continuous derivatives.
- (d) Use the second- and fourth-order finite-difference formulas to approximate the derivatives of the functions $\sin(2\pi x)$, $\cos(\pi(x-0.5))$, and $\sqrt{(1+\cos(2\pi x))^3}$ on an equispaced grid of $n = 500$ points on the periodic interval $[0, 1]$. Plot the approximation error for the derivative at each grid point and then plot the maximum absolute error on grids with $n = 100, 200, 300, \dots, 10^4$ (use a logarithmic scale for both axes). Can you explain the behavior of the error for each function (e.g., why proportional to h^2 , h^4 , etc.)?

2) Fourier series. In the Fourier basis, a 2-periodic function $f(x)$ on $[-1, 1]$ is written as

$$f(x) = \frac{1}{\sqrt{2}} \sum_{k=-\infty}^{\infty} \hat{f}_k e^{i\pi k x}, \quad \text{where} \quad \hat{f}_k = \frac{1}{\sqrt{2}} \int_{-1}^1 e^{-i\pi k x} f(x) dx.$$

- (a) Show that if f is n -times continuously differentiable with $|f^{(n)}(x)| \leq M$ on the periodic interval $[-1, 1]$, then $|\hat{f}_k| \leq \sqrt{2}M/(\pi k)^n$. (**Hint:** integrate by parts.)
- (b) If $f(x)$ is approximated by the truncated series $f_N(x) = (1/\sqrt{2}) \sum_{k=-N}^N \hat{f}_k e^{i\pi k x}$, how do you expect the approximation error $E_N = \max_{-1 \leq x \leq 1} |f(x) - f_N(x)|$ to scale as N is increased? Derive a rigorous bound for the approximation error.
- (c) Compute the Fourier coordinates of $f(x) = \sin^3(\pi x)$, $g(x) = |x|$, and $h(x) = |\sin(\pi x)|^3$. Plot the magnitude of the Fourier coefficients $-250 \leq k \leq 250$ on a logarithmic scale. Based on the coefficient plots, roughly what accuracy do you expect if you approximate g and h by truncating their Fourier series, discarding terms with $|k| > 250$? Compare your observations with your work in part (a) and (b).

Problem 3) The trapezoidal rule (or midpoint rule) approximates the integral of a function $f : [0, 1] \rightarrow \mathbb{R}$ by sampling the function on a grid and summing up its local averages:

$$I = \int_0^1 f(x) dx \approx \sum_{k=0}^{n-1} h \left[\frac{f(kh) + f(kh + h)}{2} \right] = I_n, \quad \text{where} \quad h = 1/n.$$

Complete each part below to bound the size of the approximation error, $E_n = I - I_n$, in the trapezoidal rule for integrands that are at least twice continuously differentiable on $[0, 1]$.

- (a) Use a Taylor polynomial and remainder to derive a rigorous bound for the *local approximation error* in the trapezoidal rule, which is defined (for $0 \leq a \leq 1 - h$) by

$$E^{(a)} = \int_a^{a+h} f(x) dx - h \left[\frac{f(a) + f(a+h)}{2} \right].$$

- (b) Show that $E_n = \sum_{k=0}^{n-1} E^{(kh)}$, and use the result of (a) to derive a rigorous upper bound for E_n . How does E_n behave as the number of grid points is increased, i.e., as $n \rightarrow \infty$?
- (c) If the integrand is periodic with an absolutely convergent Fourier series, given by

$$f(x) = \sum_{j=-\infty}^{\infty} \hat{f}_j e^{2\pi i j x}, \quad \hat{f}_j = \int_0^1 f(x) e^{-2\pi i j x} dx,$$

then show that the approximation error satisfies $E_n \leq \sum_{j=1}^{\infty} |\hat{f}_{jn}| + |\hat{f}_{-jn}|$.

Hint: The trapezoidal rule simplifies to $I_n = \sum_{k=0}^{n-1} h f(hk)$ when f is periodic.

- (d) If f is periodic and has $m \geq 2$ continuous, periodic derivatives on $[0, 1]$, how do you expect E_n to behave as the number of grid points is increased? What if f has $m \geq 2$ continuous derivatives but is not periodic? Explain.